

SQL: Email Campaign Report

Problem Description

An email campaign tracking platform maintains data on various campaigns and their email statistics. The task is to generate a report that lists each campaign's name along with the total number of emails sent, emails opened, and emails not opened.

The result should have the following columns:

- **campaign_name** - the name of the email campaign.
- **total_emails_sent** - the total number of emails sent in the campaign.
- **total_emails_opened** - the total number of emails opened in the campaign.
- **total_emails_not_opened** - the total number of emails not opened.

Constraints:

- The result should be sorted in **ascending order** by campaign_name.
- The number of emails not opened is calculated as the difference between the total emails sent and the total emails opened.

Database Schema

Table: campaigns

Column	Type	Constraints	Description
id	INT	NOT NULL PRIMARY KEY	The identifier of the email campaign
name	VARCHAR(255)		The name of the email campaign

Table: email_stats

Column	Type	Constraints	Description
campaign_id	INT		

		FOREIGN KEY REFERENCES campaigns(id)	The reference to the email campaign
emails_sent	INT		The number of emails sent in the email campaign
emails_opened	INT		The number of emails opened in the email campaign

Sample Data

Table: campaigns

id	name
1	SummerSale2021
2	FallPromo
3	WinterWonderland

Table: email_stats

campaign_id	emails_sent	emails_opened
1	1000	800
2	1500	1200
3	2000	1800
1	500	300
2	700	500
3	800	600
1	300	200
2	400	300
3	600	500
3	400	300

Sample Output

FallPromo	2600	2000	600
SummerSale2021	1800	1300	500
WinterWonderland	3800	3200	600

SQL Solution

```
SELECT
    c.name AS campaign_name,
    SUM(es.emails_sent) AS total_emails_sent,
    SUM(es.emails_opened) AS total_emails_opened,
    (SUM(es.emails_sent) - SUM(es.emails_opened)) AS
total_emails_not_opened
FROM
    campaigns c
JOIN
    email_stats es ON c.id = es.campaign_id
GROUP BY
    c.id, c.name
ORDER BY
    campaign_name ASC;
```